One Boundary, Two Scalings, the Same Margin

Problem. Eight emails have two features: the number of links x_1 and the number of words written in capitals x_2. Their labels are $y = +1$ (spam) and $y = -1$ (not spam, a normal email). Find a maximum-margin separating line. This example assumes **linearly separable data**.

Spam ($y = +1$)	$(3,3)$	$(4,3)$	$(3,4)$	$(4,4)$
Not spam ($y = -1$)	$(1,1)$	$(0,1)$	$(1,0)$	$(0,0)$

1. Try a separating line and calculate its scores

Choose $x_1 + x_2 = 4$. In the notation $w^\top x + b = 0$, this means $w = (1,1)^\top$, $b = -4$, and $\|w\| = \sqrt{2}$.

$$s(x) = w^\top x + b = x_1 + x_2 - 4, \qquad d_{\text{signed}}(x) = \frac{s(x)}{\|w\|}.$$

A positive score predicts spam; a negative score predicts not spam. Ordinary distance is $|d_{\text{signed}}|$.

2. Identify the nearest points, not the largest scores

Point x	Label y	Score s	ys	Ordinary distance
$(3,3)$	$+1$	2	2	$2/\sqrt{2} = \sqrt{2}$
$(1,1)$	-1	-2	2	$2/\sqrt{2} = \sqrt{2}$
$(4,3)$	$+1$	3	3	$3/\sqrt{2}$
$(0,1)$	-1	-3	3	$3/\sqrt{2}$

The remaining points have label-adjusted scores $3, 4, 3, 4$. Thus the smallest $y_i s(x_i)$ is 2, attained at $(3,3)$ and $(1,1)$. The **one-sided geometric margin** is $\rho = \min_i y_i s(x_i)/\|w\| = 2/\sqrt{2} = \sqrt{2}$.

3. Make the nearest label-adjusted score equal to 1

Divide *both* w and b by 2: $\widetilde{w} = (0.5, 0.5)^\top$ and $\widetilde{b} = -2$.

$$0.5x_1 + 0.5x_2 - 2 = 0 \iff x_1 + x_2 = 4, \qquad \|\widetilde{w}\| = 1/\sqrt{2}.$$

The nearest scores become $+1$ and -1, so their label-adjusted scores are both 1. Every other label-adjusted score is 1.5 or 2. Hence $y_i(\widetilde{w}^\top x_i + \widetilde{b}) \geq 1$ for every point. **The line has not moved:** the new margin is $1/\|\widetilde{w}\| = \sqrt{2}$, exactly as before.

4. Connect these numbers to the SVM objective

Once this scaling is fixed, maximizing the margin is equivalent to

$$\boxed{\min_{w,b} \frac{1}{2}\|w\|^2 \quad \text{subject to} \quad y_i(w^\top x_i + b) \geq 1 \quad \text{for every } i.}$$

For our candidate, the objective is $\frac{1}{2}(0.5^2 + 0.5^2) = \boxed{0.25}$. Equivalent objectives have the same optimizer; their numerical values need not be equal.

5. Why is this line actually optimal?

The opposite-class points $(3,3)$ and $(1,1)$ are $2\sqrt{2}$ apart. A line with margin ρ must place each at least ρ away on opposite sides. Their perpendicular separation is therefore at least 2ρ, but cannot exceed their distance $2\sqrt{2}$. Thus $\rho \leq \sqrt{2}$. Our line attains this bound, so it is maximum-margin. These two points are its **support vectors**. The full corridor between the margin lines is $2\rho = 2\sqrt{2}$ wide.

Takeaway: Rescaling changes scores and coefficient values, not the boundary or its distances. The constraint fixes that arbitrary scale; minimizing the norm then selects the widest margin.